Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Engineer · Agentic Systems · LangGraph · Claude · Python · C# · Enterprise Integrations · Developer Productivity

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PROFESSIONAL SUMMARY

Forward Deployed AI Engineer with 8 years building production enterprise systems and hands-on experience shipping **agentic AI solutions** directly into complex, regulated platforms. Built **Leasel** — an 8-agent LangGraph system integrating LLM orchestration, RAG, tool calling, and conditional HITL routing into a production compliance platform: the pattern of **embedding AI agents into existing enterprise workflows** and shipping them to real users. Built **AgentProbe** — an AI evaluation and regression toolkit with multi-provider LLM routing across **Claude (Anthropic), OpenAI, Groq, and Ollama**; CI-integrated golden-dataset harness; and offline + online eval loops that keep non-deterministic systems reliable in production. At Magizhs: deployed an **MCP server** integrating agentic workflow outputs into downstream tooling. At Entrata (8yr): owned enterprise integrations, APIs, and automated workflows across a multi-tenant SaaS platform in a **regulated, data-sensitive environment** — the same accountability standard this role carries. Active **Claude Code** user; Python-primary, C# (1.5yr), TypeScript; REST APIs, CI/CD, GitHub Actions, Docker/Kubernetes.

CORE COMPETENCIES

Agentic AI Frameworks | LangGraph / LangChain | Multi-Agent Orchestration | Tool Calling & Function Calling | MCP (Model Context Protocol) | Claude / Anthropic API | OpenAI / Azure OpenAI | AWS Bedrock | RAG / Knowledge Retrieval | Prompt Engineering & Structured Outputs | AI Evaluation & Regression Harnesses | LLM-as-Judge Scoring | Developer Productivity Tooling | Enterprise Platform Integration | REST APIs & API Design | Python / FastAPI | C# / .NET (1.5yr) | TypeScript / JavaScript | Java / Spring Boot | CI/CD (GitHub Actions) | Docker / Kubernetes | Workflow Automation | HITL / Human-in-the-Loop | Multi-Tenant Architecture | Regulated / Data-Sensitive Environments

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building AI-enabled applications and data pipelines; integrating AI agents and LLM components into backend systems; surfacing business insights into application-facing APIs used by internal engineering teams.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic workflow system** translating natural-language requests into structured outputs — LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) with **MCP server deployment** for downstream tool integration; integrated with enterprise operational tooling (MAVLink/QGroundControl).
- Implemented a **RAG layer** over FAA regulatory corpus with cited-source grounding; enforced **validation guardrails** blocking non-compliant outputs before export — every compliance check grounded in source regulation.

Entrata — PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Owned and shipped enterprise integrations and automated workflows** across a complex multi-tenant platform — compliance pipelines, document management, and async processing systems handling 100K+ monthly active users; 80% reduction in manual operational workload via Python/Bash automation.
- **Built and maintained APIs and backend services** (Python, PHP, Java, PostgreSQL, RabbitMQ) across 120+ Agile sprints; led microservices redesign (Docker) — 50% scalability improvement, 30% deployment time reduction.
- **Regulated, data-sensitive environment:** 100% PHI compliance, Libsodium-encrypted pipelines, multi-tenant data isolation at the schema layer; zero security incidents across full AWS S3 migration.
- Optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers; conducted 300+ code reviews; shipped 20+ production modules end-to-end.

Entrata India — Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built compliance and document management modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration; income-limits import (1.6M HUD/Tax Credit records across 5 years).

PROJECTS

Leasel: 8-Agent Agentic AI Platform - Enterprise Workflow Integration

LangGraph · MCP · Agentic AI · Developer Productivity · RAG

Production agentic AI system embedded into a compliance engineering workflow: **8-agent LangGraph StateGraph** (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with **tool calling, durable state, conditional routing, and HITL gates**. Integrates heterogeneous enterprise data sources — product knowledge, regulatory rules, historical incidents, system configuration — into a single reasoning pipeline. **Core FDE pattern:** embedded directly into an existing engineering workflow, replacing a manual multi-hour investigation process; engineers interact through a structured API surface, not a chatbot. RAG: hybrid TF-IDF + semantic reranking with citation grounding to exact source clauses. Eval: 93.6% precision on 1,240 test cases; CI regression gate.

Python · LangGraph · FastAPI · MCP · TF-IDF · ChromaDB · OCR · SQLite · GitHub Actions · 2026

AgentProbe: AI Evaluation Toolkit — Multi-Provider LLM Routing

Claude · OpenAI · Groq · Ollama · Eval Harness · CI/CD

Open-source toolkit for evaluating and regression-testing AI agents across providers: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **LLM-as-judge scoring**, CI-integrated regression harness with golden datasets, and offline + online evaluation loops. Multi-provider routing across **Claude (Anthropic), OpenAI, Azure OpenAI, Groq, and Ollama** — the same providers this role targets. Prompt version management across providers; reusable evaluation components deployed in CI/CD via GitHub Actions. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · Claude API · OpenAI API · CI/CD (GitHub Actions) · 2025

AI on FHIR: NLP-Driven Healthcare API Integration

LLM Integration · REST APIs · Enterprise Data · HIPAA

NLP platform integrating LLMs with enterprise healthcare APIs: translated natural-language queries into **FHIR API calls** using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across structured medical records. **~60% reduction in manual clinical retrieval time**; bilingual (EN/ES). Demonstrates the core FDE pattern: LLM-powered interface layered over an existing enterprise data API without replacing it.

Python · Flask · React · LLMs · BERT · FHIR APIs · Docker · spaCy · 2024

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU, Anantapur 2017